

ALI JABBAR

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SUMMARY

Computer Science graduate (CGPA 3.93/4.00; 100% merit scholarship) who builds reliable Python systems across forecasting, OCR, browser automation, recommendation algorithms, and LLM integration. Strong in OOP, SQL, software design, validation, debugging, and REST integration; turns ambiguous workflows into modular, explainable components with clear failure handling. Teaching Assistant who mentored 30+ students through code reviews and technical problem-solving.

EDUCATION

University of Central Punjab — Lahore, Pakistan

Oct 2022 – Jul 2026

- Bachelor of Science in Computer Science
- CGPA: 3.93/4.00 | 100% Merit Scholarship
- AWS Academy Graduate — Cloud Foundations (EC2, S3, Cloud Architecture), 2024

WORK EXPERIENCE

University of Central Punjab

Oct 2024 – May 2025

Teaching Assistant, Object-Oriented Programming

- Mentored 30+ students through weekly code reviews, debugging sessions, and 1:1 design guidance on inheritance, polymorphism, abstraction, and clean code.
- Reviewed submissions to surface recurring implementation and design issues, then converted those patterns into targeted explanations and refactoring guidance.

PROJECTS

SEBPS — Smart Electricity Bill Prediction & Advisory System

Collaborative Final-Year Project | Owned modules: Python, forecasting, OpenCV/OCR, Playwright, BeautifulSoup, Groq API | Integrated platform: Django REST, TimescaleDB, React, Docker Compose

- Designed and implemented five intelligence and automation modules spanning forecasting and tariff calculation, OCR/reference extraction, 12-month LESCO history collection, appliance/budget recommendations, and a grounded AI advisor; integrated them into a teammate-built full-stack platform.
- Built a multi-model forecasting pipeline using Holt-Winters, EWMA, WMA, seasonal naive, and trend models; selected the best consumer-specific model with LOOCV/MAE and blended it with live meter projections through sigmoid confidence weighting.
- Engineered OpenCV preprocessing, Tesseract/EasyOCR fallback, regex normalization, and majority voting; automated portal navigation and parsing with Playwright retries, BeautifulSoup, confidence scores, diagnostics, and graceful failures.

Movie Recommendation System

Python, Pandas, NumPy, scikit-learn, TF-IDF, cosine similarity, React 18, Vite, Tailwind CSS, Framer Motion

- Built an interpretable content-based recommender over the TMDb 5000 dataset by combining overview, genre, and keyword features, applying TF-IDF with 5,000 features, and ranking movies by cosine similarity.
- Precomputed the top six recommendations per movie into JSON for fast client-side delivery, then built a React/Vite interface—avoiding backend complexity for a static inference workload.

Binary Classification of Diabetes — ANN from Scratch

Python, NumPy, Pandas, Matplotlib, Jupyter Notebook | Neural network implemented without TensorFlow or PyTorch

- Implemented a 9–4–1 binary classifier from first principles in NumPy, including forward propagation, backpropagation, sigmoid activation, and gradient descent.
- Standardized eight BRFSS 2015 health features, injected a bias unit, trained for 300 epochs at a 0.1 learning rate, and monitored MSE and accuracy curves to verify learning behavior.

LEADERSHIP ACTIVITIES

IEEE Robotics and Automation Society Student Chapter — Founding Member

Oct 2024 – Oct 2025

- Co-founded the chapter and organized hackathons, workshops, and technical events to expand peer learning and hands-on engineering practice.

SKILLS

Programming: Python, Java, C/C++, JavaScript, SQL

Software Engineering: OOP, algorithms, analytical problem-solving, modular design, REST APIs, input validation, error handling, debugging, code review, Git/GitHub

ML & Data: NumPy, Pandas, scikit-learn, feature engineering, NLP, TF-IDF, cosine similarity, time-series forecasting, model evaluation (LOOCV, MAE)

AI, CV & Automation: OpenCV, Tesseract, EasyOCR, Playwright, BeautifulSoup, regular expressions, Groq API, prompt/context engineering

Platform & Cloud: AWS (EC2, S3); integration exposure to Django REST Framework, Celery, PostgreSQL/TimescaleDB, React/Vite, Docker Compose

Languages: English (Fluent), Urdu (Native), Arabic (Conversational)